

4.4. Prove that the following formula is a tautology.

$$U_1 = \underbrace{(p \wedge q \rightarrow r)}_A \rightarrow \underbrace{(p \rightarrow (q \rightarrow r))}_B$$

Theoretical results

1) U is a tautology ($\models U$)
if $(\forall i, i(U) = T)$

2) U is consistent if
 $\exists i, i(U) = T, i$ -model.

3) U is inconsistent if
 $(\forall i, i(U) = F)$.

4) U is contingent if
 $\exists i, i(U) = T, i \rightarrow$ model of U .
 $\exists j, j(U) = F, j \rightarrow$ anti-model of U .

	p	q	r	$p \wedge q$	A	$q \rightarrow r$	B	U_1
i1	T	T	T	T	T	T	T	T
i2	T	T	F	T	F	F	F	T
i3	T	F	T	F	T	T	T	T
i4	T	F	F	F	T	F	T	T
i5	F	T	T	F	T	T	T	T
i6	F	T	F	F	T	F	T	T
i7	F	F	T	F	T	T	T	T
i8	F	F	F	F	T	F	T	T

Conclusion: for $(\forall i)_m, i(U) = T$, so $U \rightarrow$ valid formula, tautology.

② Using the truth ... decide the type of the formula.

$$U_1 = \underbrace{q \vee \neg p \vee r}_A \rightarrow \underbrace{\neg(p \vee r)}_B$$

	p	q	r	$\neg p$	$p \vee r$	B	A	U_1
i1	T	T	T	F	T	F	T	F
i2	T	T	F	F	T	F	T	F
i3	T	F	T	F	T	F	T	F
i4	T	F	F	F	T	F	T	F
i5	F	T	T	T	T	F	T	F
i6	F	T	F	T	F	T	T	T
i7	F	F	T	T	T	F	T	F
i8	F	F	F	T	T	T	T	T

$\Rightarrow U_1$ contingent formula having 3 models (i4, i6, i8)

model i4: $\{p, q, r\} \rightarrow \{T, F\}$

$i_4(p) = T, i_4(q) = F, i_4(r) = T, i_4(U_1) = T$.

anti-model i1:

i1: $\{p, q, r\} \rightarrow \{T, F\}$

$i_1(p) = T, i_1(q) = T, i_1(r) = T, i_1(U_1) = F$.

Ex. 5.4 Find DNF/CNF ; valid formulas?

$$U_4 = (p \rightarrow (q \vee r)) \rightarrow ((p \rightarrow q) \vee (p \rightarrow r))$$

1) Replace the " $\rightarrow$ ".

$$\stackrel{1,3}{=} \neg p \vee (q \vee r) \rightarrow (\neg p \vee q \vee \neg p \vee r)$$

$$\stackrel{2}{=} \neg(\neg p \vee q \vee r) \vee (\neg p \vee q \vee \neg p \vee r)$$

De Morgan

$$\equiv (p \wedge \neg q \wedge \neg r) \vee (\neg p \vee q \vee \neg p \vee r)$$

$$\equiv (\underline{p \wedge \neg q \wedge \neg r}) \vee (\neg p \vee q \vee r)$$

$\Rightarrow$ DNF with 4 cubes.

Distrib.

$$\equiv p \vee (\neg p \vee q \vee r) \wedge (\underline{\neg q} \vee \neg p \vee q \vee r) \wedge (\neg r \vee \neg p \vee q \vee r)$$

$\Rightarrow$ CNF with 3 clauses.

$$\equiv T \wedge T \wedge T \equiv T. \Rightarrow \text{valid formula.}$$

$$\text{DNF: } \vee (\wedge)_{i,j}$$

$$\text{CNF: } \wedge (\vee)_{i,j} ?$$

$$p \vee \neg p \vee q \equiv T$$

$$p \wedge \neg p \wedge r \equiv F$$

$$A \rightarrow B \equiv \neg A \vee B$$

$$A \cdot A \equiv A$$

$$\neg p \vee \neg p = \neg p$$

Idempotency law
?

5.2

6.1. Determine the models of the formula using DNF.

$$U_1 = (p \vee q \rightarrow r) \rightarrow (p \rightarrow r) \wedge q$$

replace 1,2,3

$$\equiv \neg(\neg(p \vee q) \vee r) \vee (\neg p \vee r) \wedge q$$

De Morgan

$$\equiv (p \vee q \wedge \neg r) \vee ((\neg p \vee r) \wedge q)$$

Distrib.

$$\equiv (p \wedge \neg r) \vee (q \wedge \neg r) \vee (\neg p \wedge q) \vee (r \wedge q)$$

$\Rightarrow$ DNF with 4 cubes.

Cube $p \wedge \neg r$ provides 2 models:

$$\hookrightarrow i_{1,2} : \{p, q, r\} \rightarrow \{T, F\}$$

$$i_1(p) = T, i_1(q) = T, i_1(r) = F$$

$$i_2(p) = T, i_2(q) = F, i_2(r) = F$$

Cube $q \wedge \neg r$ provides 2 models:

Precedence: $\rightarrow$
 $\neg, \wedge, \vee$
 $\cdot, +$

$$\hookrightarrow i_{3,4}: \{p, q, r\} \rightarrow \{T, F\}$$

$$i_3(p) = T \quad i_3(q) = T \quad i_3(r) = F$$

$$i_4(p) = F \quad i_4(q) = T \quad i_4(r) = F$$

Cube $7p \wedge q$ provides 2 models:

$$\hookrightarrow i_{5,6}: \{p, q, r\} \rightarrow \{T, F\}$$

$$i_5(p) = F \quad i_5(q) = T \quad i_5(r) = F$$

$$i_6(p) = F \quad i_6(q) = T \quad i_6(r) = T$$

Cube $r \wedge q$ provides 2 models:

$$\hookrightarrow i_{7,8}: \{p, q, r\} \rightarrow \{T, F\}$$

$$i_7(p) = F \quad i_7(q) = T \quad i_7(r) = T$$

$$i_8(p) = T \quad i_8(q) = T \quad i_8(r) = T$$

$$i_1 = i_3$$

$$i_4 = i_5$$

$$i_6 = i_7$$

$\Rightarrow U$ has 5 models: i_1, i_2, i_4, i_6, i_8 , $(U) = T$.

8.1 Write all the anti-models of the following using CNF.

$$U_1 = (q \wedge r \rightarrow r) \rightarrow (p \rightarrow r) \wedge q$$

Absorption law

$$U_1 = (\neg p \vee r) \wedge q$$

$\Rightarrow$ CNF with 2 clauses.

Clause $\neg p \vee r \equiv F$ provides 2 anti-models:

$$\hookrightarrow i_{1,2}: \{p, q, r\} \rightarrow \{T, F\}$$

$$i_1(p) = T, i_1(q) = T, i_1(r) = F$$

$$i_2(p) = T, i_2(q) = F, i_2(r) = F$$

Clause q provides 4 anti-models:

$$\hookrightarrow i_{3,4,5,6}: \{p, q, r\} \rightarrow \{T, F\}$$

$$i_3(p) = \begin{matrix} T \\ T \\ F \\ F \end{matrix} \quad i_3(q) = F \quad i_3(r) = \begin{matrix} T \\ F \\ T \\ F \end{matrix}$$

$i_2 \approx i_4 \Rightarrow U_1$ has 5 anti-models